VVS Hospital — What's Done and What's Pending

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Summary

All eight build phases (0–7) of VVSDesk are finished and running on the demo site; what remains are owner decisions, keys and go-live setup, plus some optional extra testing.

Measure	Now
Build phases done	8 of 8 (Phase 0 Foundation → Phase 7 Hardening)
Backend automated tests	114 passing
Browser (E2E) tests	23 passing, in Google Chrome
Live audits (acting as real users)	Lab 81/81 · OPD 61/61 · IPD & ER 70/70 · Security 97/98 (1 owner warning: 2FA off)
Urdu coverage of screen text	2,301 of 2,301
Branches live	1 (Lahore Main)
Decisions recorded	D-01 … D-56 in `docs/DECISIONS.md`

The app is `hospital_vvs` on ERPNext v16; staff use the React app at `/hms`, with the pharmacy POS, lab, OPD, IPD, ER, cash and finance screens all inside it.

Done: the eight build phases

Every phase in the build plan is complete, with its acceptance workflows (W1–W18) passing as automated tests.

Phase	What it delivered	Workflows
0 Foundation	Roles, Clinic Settings, React app (English / Urdu / Arabic, right-to-left), login with 2FA step, idle lock screen, access log, print agent, encrypted backups, health check	Login, role home
1 Patients, OPD, messaging	MR numbers and patient card, CNIC (masked), duplicate check and merge, households, doctors and slots, web + WhatsApp booking, OPD slip and queue, prescriptions with allergy / interaction alerts, signed and amended records	W1 (OPD), W2, W2a, W9, W10, W11
2 Pharmacy, lab, cash	Medicines by batch and expiry, pharmacy POS, customer displays, pick → review → handover, lab samples, results, verification, critical values, outsourced tests, WhatsApp report link, cashier shifts on every counter	W1, W3, W6a, W6b, W6c, W7, W14
3 IPD and ER	Wards and live bed map, admission to discharge, MAR, diet, procedures with consent, packages, IPD bill, Emergency board and triage, nurse roster	W4, W18
4 Online visit	Public request form, status page, staff dashboard, video link, fees and refunds	W5
5 Patient comms + feedback	Nursing stations, bedside page with voice-to-text, broadcasts, feedback surveys	W6, W16
6 Finance	Panels and insurance (pre-auth, claims), doctor share and payouts, welfare / zakat funds, expenses with approvals, purchasing, MIS reports	W8, W11, W12, W13

Phase	What it delivered	Workflows
7 Hardening	Security audit and 12 fixes, field-level hiding, branch lock, password-guess limits, version history, performance checks, staging anonymizer, one-command demo seeder	W15, W17

Done: work added on request

Beyond the plan, these were built at the owner's request (newest first; decision numbers in `docs/DECISIONS.md`).

Date	Item	Decision
11 Nov 2026	IPD & ER live audit: refund before discharge, 100 % insurer cover, vitals sanity, ER identify fixed	D-56
11 Nov 2026	Urdu on every screen, checked automatically in CI	D-55
11 Nov 2026	Browser tests (Playwright) and recurring expenses (rent, utilities)	D-54
11 Nov 2026	Security hardening audit with 12 fixes	D-53
11 Nov 2026	Prescription reprint from history, previous list and queue	—
10 Nov 2026	OPD flow with nurse vitals: token → vitals → doctor → done	D-52
10 Nov 2026	English → Urdu name typing, auto prescription quantity, pending patients, daily demo script	D-51
10 Nov 2026	Lab report print: QR verification, logo, digital signature line, previous results	D-50
10 Nov 2026	Lab audit, priority tags, day / week / month filters	D-49
10 Nov 2026	Handwritten / outside lab reports, print and hand over	D-48
10 Nov 2026	Card gateway (mock mode), FBR day-end reporting (mock), 38 WhatsApp templates	D-47

Date	Item	Decision
10 Nov 2026	Islamabad branch removed; single branch Lahore Main	D-45
10 Nov 2026	Doctor commission (doctors as suppliers, earned when paid)	D-43
10 Nov 2026	Pharmacy POS polish: paper prescriptions, cash in / out, invoice cards, customer display	D-42
10 Nov 2026	Branch and counter lock via POS Profile users	D-41
9 Nov 2026	Pharmacy POS rebuilt in Restaurant OS style	D-40
9 Nov 2026	Contract pharmacy as separate company; reception counter with offline queue	D-38, D-39

All demo users' passwords were reset to the shared demo password (Administrator unchanged).

Quality: tests and audits

Four live audits, run as the real demo users, found 21 defects; all are fixed and covered by regression tests.

Audit	Result	Main defects fixed	Report
Lab QA	81 / 81	Reject after result entry, lab user could amend verified results, refund did not cancel tests	`docs/QA-LAB.md`
OPD QA	61 / 61	Blank prescription quantity sold as 1 unit, shift close counted a typed total as zero	`docs/QA-OPD.md`
Security (Phase 7)	97 / 98	Clinical data reachable by cashiers / finders, branch not checked on 39 endpoints, unlimited password guesses, 17 record types without history, patients hidden across branches	`docs/QA-HARDENING.md`

Audit	Result	Main defects fixed	Report
IPD & ER QA	70 / 70	Discharge without refunding extra advance, final bill crash at 100 % insurer cover, impossible vitals accepted, ER identify failed for admin	`docs/QA-IPD.md`
Pharmacy QA	done 9 Nov	Pay-at-handover flow and demo clean-up	`docs/QA-PHARMACY.md`

Automated checks:

- **114 backend tests** cover every workflow W1–W18 with accounting checks (GL balances, payer split, doctor share).
- **23 browser tests** walk the real screens: OPD from registration to signed prescription, security, branch lock, POS, lab, public pages, recurring expenses, Urdu.
- **Performance targets** all met: patient search 3–7 ms, POS item add 30 ms, lab PDF 0.6 s, app load 54 ms (targets 300 ms – 3 s).
- **Urdu check** fails the build if new screen text has no translation.

Pending: owner decisions before go-live

These need the clinic owner's choice, keys or servers; the system is ready to take each one.

Security settings

- [ ] Switch on two-factor login for Admin, Accountant, Cash Supervisor and Pathologist (Clinic Settings → Enforce two-factor; System Settings → Enable 2FA)
- [ ] Shorten desk sessions from 170 hours to 12 hours
- [ ] Tighten login lockout from 10 tries / 60 s to 5 tries / 300 s
- [ ] Replace the shared demo password with real staff passwords

Keys and providers

- [ ] Choose the card payment gateway and add its keys (mock mode today)
- [ ] FBR POS ID, NTN, token and which bills are reported, from the tax advisor (mock mode today)
- [ ] Submit the 38 WhatsApp templates (`docs/whatsapp/`) to Meta, incl. `visit_feedback` ; choose the SMS provider

- [ ] Video service for online visits (Jitsi today) and the real online fee and wallet numbers

Setup and data

- [ ] Upload the clinic logo and the pathologist's signature; set signatory name, PMDC no. and licence line on lab reports
- [ ] Create `staging.hospital` for the full restore drill (needs the MariaDB root password) and a `training.hospital` site
- [ ] Hosting choice: on-premise server with off-site backups, or cloud with backup internet
- [ ] Real rates: wards, ICU, nursing, diet, procedures; discount limits; drug-interaction reference list
- [ ] Import real patients, doctors, medicine stock and opening balances (data manager is ready)
- [ ] Legal advice on Pakistani data-protection duties; zakat / welfare eligibility rules
- [ ] Bedside tablets per room and which wards' nursing stations to staff
- [ ] OK to remove old Phase 2 test lab orders on MR-LHR-000057 (LAB-2026-00009 … 00022)

Pending: optional technical work

None of this blocks go-live; each item can be done without the owner.

Item	Why	Size
Browser tests for IPD admission → discharge (W4), online visit (W5), pharmacy pick → review → handover (W6b), shift close (W7)	Backend tests cover them; browser tests would catch screen breakages	Medium
Arabic translation	128 of 2,301 screen texts; the rest shows in English	Medium
Native-speaker review of the Urdu wording (medical and finance terms)	Translations were written by the developer	Small (owner's staff)
Silent printing tested on real thermal / label printers (QZ Tray)	Tested through the browser print dialog only	Small, needs hardware

Item	Why	Size
Error tracking (Sentry) and uptime / SSL monitoring	Hooks documented; needs a DSN and monitoring host	Small
CI on a remote repository	Workflow file is written; no remote repo connected	Small
Lab analyzer interface	Optional; only if an analyzer with LIS output is bought	Large
Recurring daily demo data in cron (`scripts/opd_demo.sh`)	Not scheduled; the crontab line is in the script	Tiny
Repository push to GitHub	Needs the owner's sign-in	Tiny

Known limits today: a few Urdu sentences built from pieces read in English word order; switching language clears half-typed forms; the ICU has no free bed because audit runs left patients admitted.

Where things are

What	Where
Staff app (React)	`http://hospital.localhost:8011/hms`
ERPNext Desk	`http://hospital.localhost:8011/desk` (VVS Hospital workspace)
Demo logins	`<name>@vvs.demo` — reception, nurse, dr.ayesha, cashier, lab, pathologist, accounts, supervisor, admin …
App code	`apps/hospital_vvs` (backend) and `apps/hospital_vvs/frontend` (React)
Progress and decisions	`docs/PROGRESS.md`, `docs/DECISIONS.md` (D-01 … D-56)
Operations (backups, restore drill, monitoring, hardware)	`docs/OPS.md`
API reference	`docs/API.md` (every whitelisted method)

Useful commands (run from the bench folder):

- Backend tests: `bench --site hospital run-tests --app hospital_vvs`
- Browser tests: `cd apps/hospital_vvs/frontend && npm run e2e`
- Urdu coverage: `npm run i18n:check`
- Live audits: `bench --site hospital execute hospital_vvs.devtools.qa_hardening.run` (also `qa_lab`, `qa_opd`, `qa_ipd`)
- Fresh demo / training data: `bench --site <site> execute hospital_vvs.setup.demo.seed_all`
- A day of OPD demo traffic: `apps/hospital_vvs/scripts/opd_demo.sh`